① $\frac{\partial^2 n_e'}{\partial t^2} - C_s^2 \nabla^2 n_e' = \frac{n_0 Z e^2}{m_e m_i c^2} \nabla^2 (\vec{A}_L \cdot \vec{A}')$.

$\vec{A}_L = \frac{\vec{A}_L}{2} [\exp(i\vec{k}_0 \cdot \vec{x} - i\omega_0 t) + \exp(-i\vec{k}_0 \cdot \vec{x} + i\omega_0 t)]$

$\vec{A}' = \vec{A}' \exp(-i\vec{k} \cdot \vec{x} + i\omega t)$, $n_e' = n_e' \exp(-i\vec{k}\vec{x} + i\omega t)$.

$$(-\omega^2 + C_s^2 k^2) n_e' = \frac{n_0 Z e^2}{m_e m_i c^2} \nabla^2 \left[\frac{\vec{A}_L}{2} \cdot \vec{A}' \left(e^{-i(k-k_0)x + i(\omega-\omega_0)t} + e^{-i(k+k_0)x + (\omega+\omega_0)t}\right)\right]$$

$\nabla^2 = -k^2$.

$$(\omega^2 - C_s^2 k^2) n_e' = \frac{Z n_0 e^2}{m_e m_i c^2} k^2 \frac{\vec{A}_L}{2} \cdot [\vec{A}'(k-k_0, \omega-\omega_0) + \vec{A}'(k+k_0, \omega+\omega_0)]$$

(k_0, k_s)

$$(\omega^2 - C_s^2 k^2) n_e' = \underbrace{\frac{n_0 Z e^2 k_s^2}{m_e m_i c^2}}_{\beta} A_L A'$$

$$(\omega^2 - c^2 k_s^2 + \omega_{pe}^2) A' = \underbrace{\frac{4\pi e^2}{m_e}}_{\alpha} n_e' A_L$$

$$(\omega^2 - C_s^2 k^2) n_e' = \beta \frac{\alpha n_e' A_L^2}{\omega^2 - c^2 k_s^2 + \omega_{pe}^2}$$

$$A_L^2 = \frac{(\omega^2 - C_s^2 k^2)(\omega^2 - c^2 k_s^2 + \omega_{pe}^2)}{\alpha\beta}$$

$$(\omega^2 - C_s^2 k^2)(\omega^2 - c^2 k_s^2 + \omega_{pe}^2) = \alpha\beta A_L^2$$